The Art and Science of Transportation Research in the AI Era

Git

Meng Cai



A stray
/ə streɪ/

Ashtray
/ˈæʃ treɪ/

Learning goals

#1 Become familiar with Git and GitHub

#2 Explain key Git terminology

#3 Clone the course repo

Agenda

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#1 What is Git

#2 Why use Git

#3 How to use Git

#4 AI reimbursement

#5 Present your research proposal ideas

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#1 What is Git

#2 Why use Git

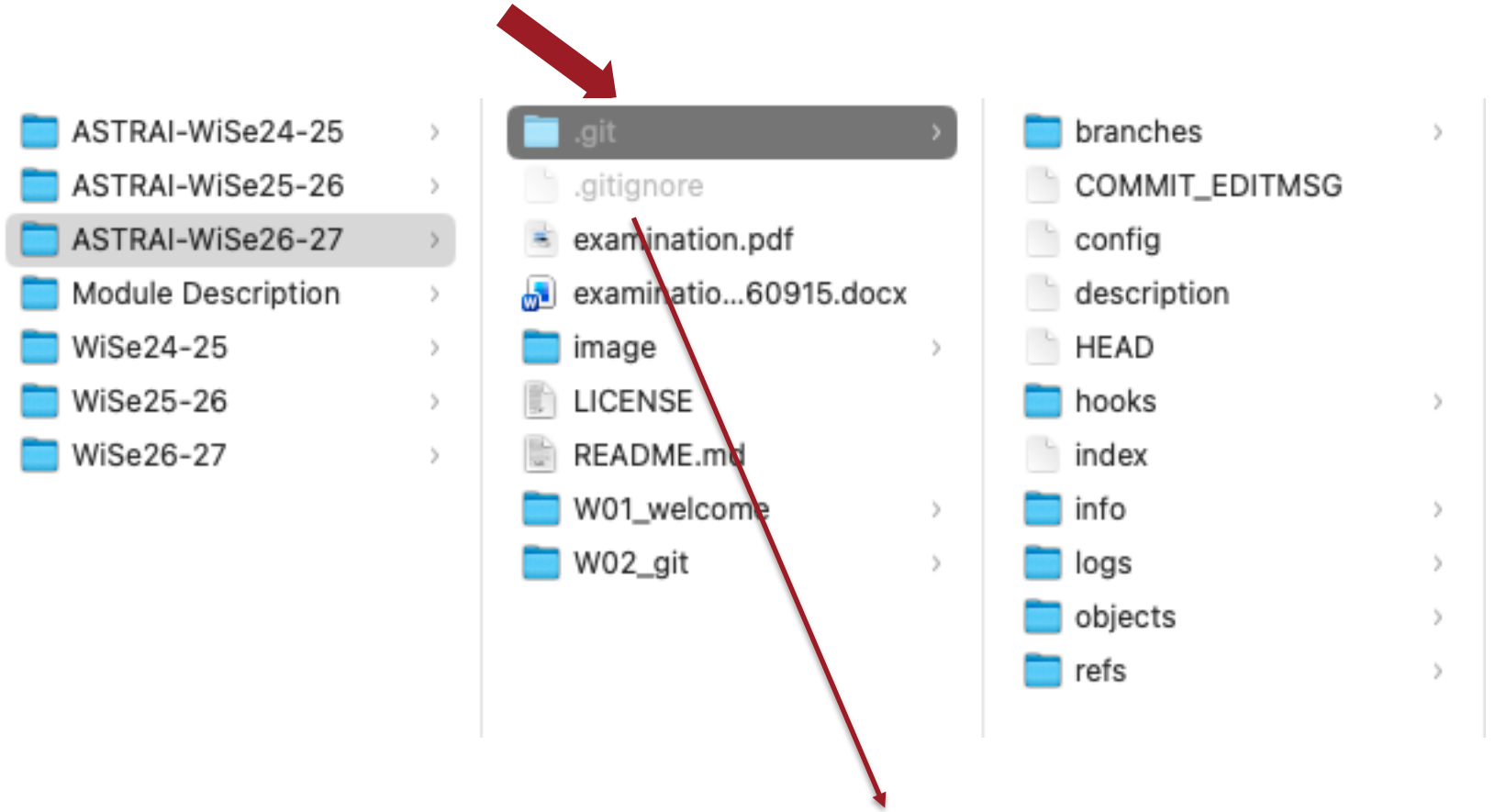
#3 How to use Git

#4 AI reimbursement

#5 Present your research proposal ideas

#1.1 What is Git

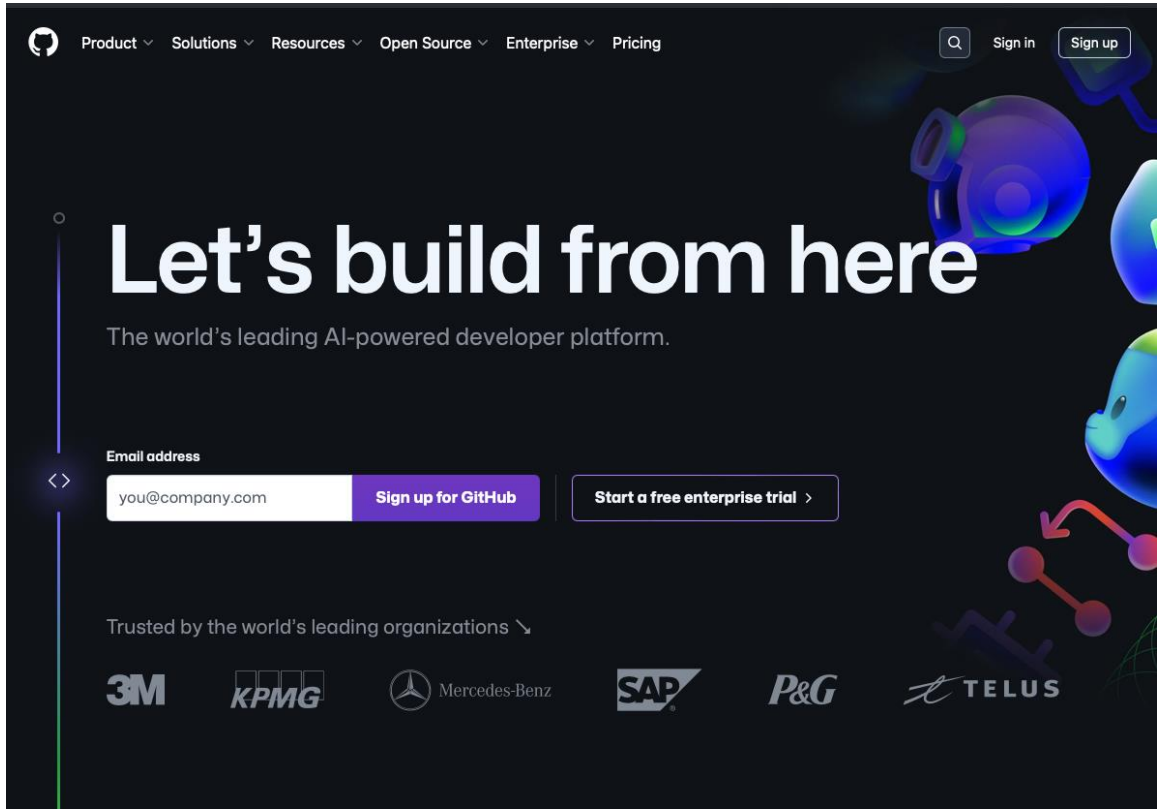
- Git is a free tool that records the history of your project files.
- Every saved version becomes a snapshot, called a *commit*.
- Git runs on your computer and does the version tracking.



A list of files and folders that Git should **not** track

#1.2 What is GitHub

- GitHub is a website that stores git projects online.
- It works as a backup and a place to share your project.
- Others can download your project and contribute to it.



#1.2 What is GitHub

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Commits

main | All users | All time

Commits on Oct 14, 2026

- week 1 slides — caimeng2 committed last week — 30bbfa8
Commits on Sep 21, 2026
- minor change wording — caimeng2 committed last month — 6a67230
Commits on Sep 18, 2026
- w01 reading — caimeng2 committed on Sep 18 — 574ba15
Commits on Sep 16, 2026
- schedule Morteza — caimeng2 committed on Sep 16 — 5f87822
- add license — caimeng2 committed on Sep 16 — 99562c9
Commits on Sep 15, 2026
- add exam file — caimeng2 committed on Sep 15 — 546a76b
Commits on Sep 14, 2026
- remove backticks — caimeng2 committed on Sep 14 — 3150a99

Commit 3150a99

caimeng2 committed on Sep 14

remove backticks

main

1 parent 560c4fd commit 3150a99

1 file changed +1 -1

Search within code

README.md

```
@@ -50,7 +50,7 @@ The schedule and topics may change based on class progress and interests.

 Week 1 (Oct 14, 2026): Welcome to ASTRAI

-- Week 2 (Oct 21, 2026): `git`; presenting your initial ideas for YOU-lead
+- Week 2 (Oct 21, 2026): Git; presenting your initial ideas for YOU-lead

 Week 3 (Oct 28, 2026): Traffic flow simulation ([Dr.-Ing. Wei Jiang](https://www.linkedin.com/in/wei-jiang-52083a32/))
```

#2.1 Why use Git

Without Git	With Git
analysis.py	analysis.py
analysis_v2.py	change method
analysis_final.py	add data
analysis_final_new.py	clean data
analysis_final_FINAL.py	add plot

- Keep one file and a full history
- Undo mistakes
- Go back to an earlier version

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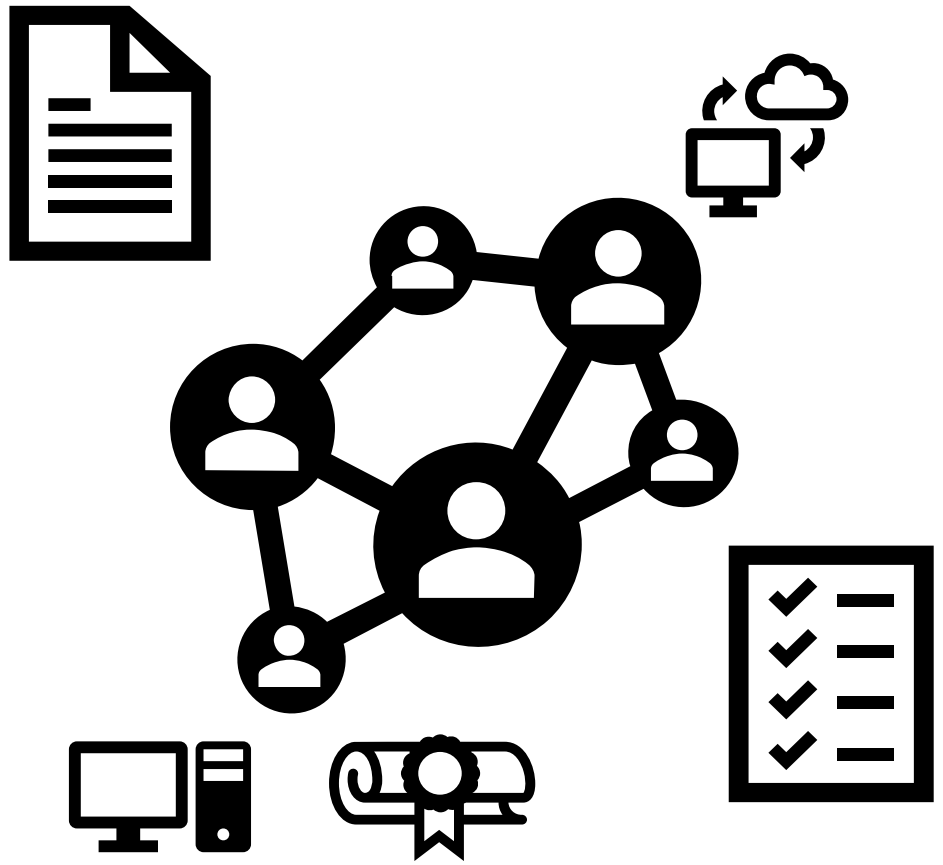
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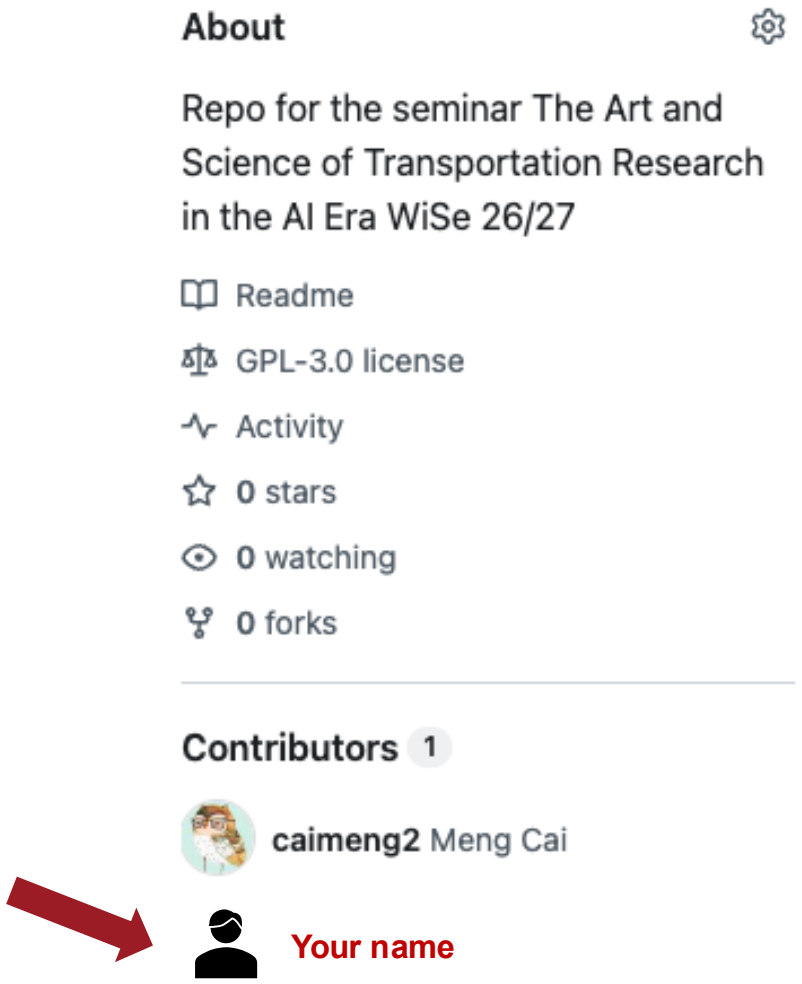
#2.2 Why use GitHub

- Backup online and access anywhere
- Collaborate: share code and data without emailing files
- Reproducible research: others can see exactly which version produced your results
- Open research: find and reuse code from others worldwide (more in Week 5)



#2.3 Why we use Git/GitHub

- Git is an invaluable tool for anyone working with code.
- If you use AI to edit your code, Git shows exactly what changed and lets you undo it.
- Research labs and industry also use Git.
- It keeps course materials open, backed up, and accessible from any device.
- It allows us to build this course together.



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#3.1 Git terminology

Repo: A git repository is a folder or directory that contains all the files and the complete history of changes for a project. There can be a local repository (on your computer) and a remote repository (hosted on GitHub)
Clone: To create a copy of a remote repo on your local machine.
Fork: Forking creates a copy of a repo on a remote platform such as GitHub. This copy is independent of the original and you can make changes to it without affecting the original project.
Commit: A commit is a snapshot of the changes made to the files in a repo at a specific point in time.
Push: When you want to upload your local changes to the remote repository, you push them.
Pull: To update your local repo with changes from a remote repo, you pull those changes. This is typically done before pushing your own changes to ensure you have the latest update.
Pull request: A pull request is a way to propose changes to a repository. It allows collaborators to review and discuss the proposed changes before merging them into the main branch.

Adapted from: https://medium.com/@unaware_harry/all-about-git-terminology-d9c91ea0a734
More info: https://git-scm.com/docs/gitglossary

#3.2 Common git commands

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Command	Function	Syntax
git init	starts a new repository	git init
git clone	duplicates a remote repository to your local machine	git clone <repository-url>
git status	shows the current status of the working directory and staging area	git status
git add	adds files to the staging area, preparing them for commit	git add <file-name> git add . # Add all files
git commit	commits changes in the staging area to the repository, recording a snapshot of the project history	git commit -m "commit message"

#3.2 Common git commands

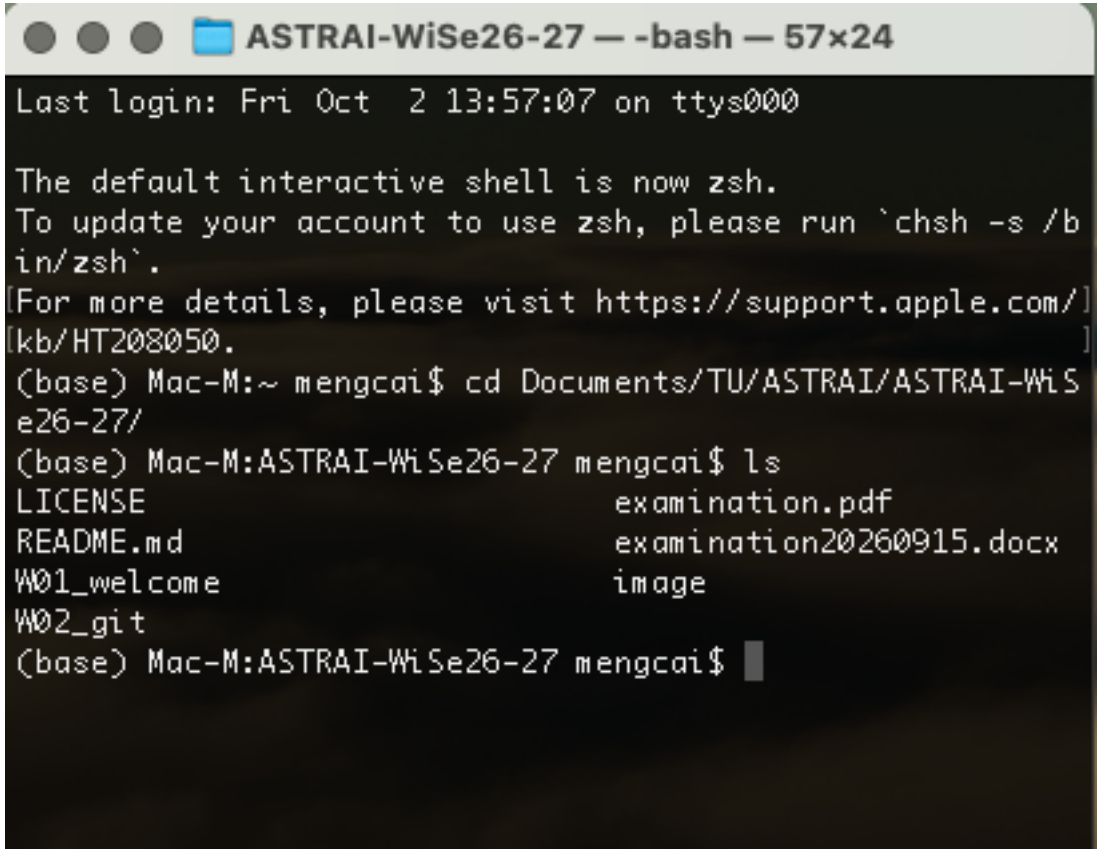
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Command	Function	Syntax
git push	pushes changes to a remote repository	git push origin <branch-name>
git pull	fetches changes from a remote repository and merges them into the current branch	git push origin <branch-name>
git branch	lists all branches or creates a new branch	git branch # List all branches git branch <new-branch-name> # Create a new branch
git checkout	switches to another branch or commit	git checkout <branch-name>
git merge	merges a branch into the current branch	git merge <branch-name>

#3.3 How to interact with Git

- Command line (Terminal on Mac, Git Bash on Windows), where you type commands
- GitHub Desktop, where you click buttons instead of typing
- Code editor (e.g., VS Code, Atom, Cursor), which has Git built in
- GitHub website, where you browse, edit, and download files in your browser



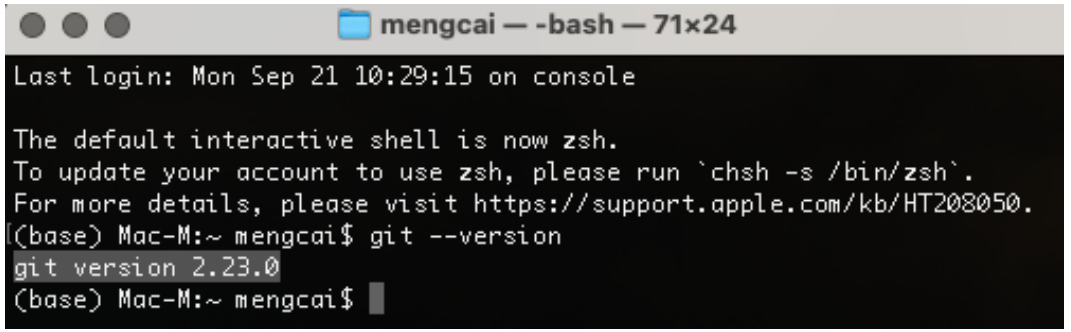
#3.4 Install Git

- **Instructions**: https://github.com/git-guides/install-git

- **Check if Git is installed**: open Terminal (Mac/Linux) or Git Bash (Windows) and type:

  git --version

  You should see a version number
  The number itself does not matter



#3.5 Clone ASTRAI repo

Do you still remember what **git clone** means?

#3.5 Clone ASTRAI repo

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What's the difference between **clone** and **pull**?

#3.5 Clone ASTRAI repo

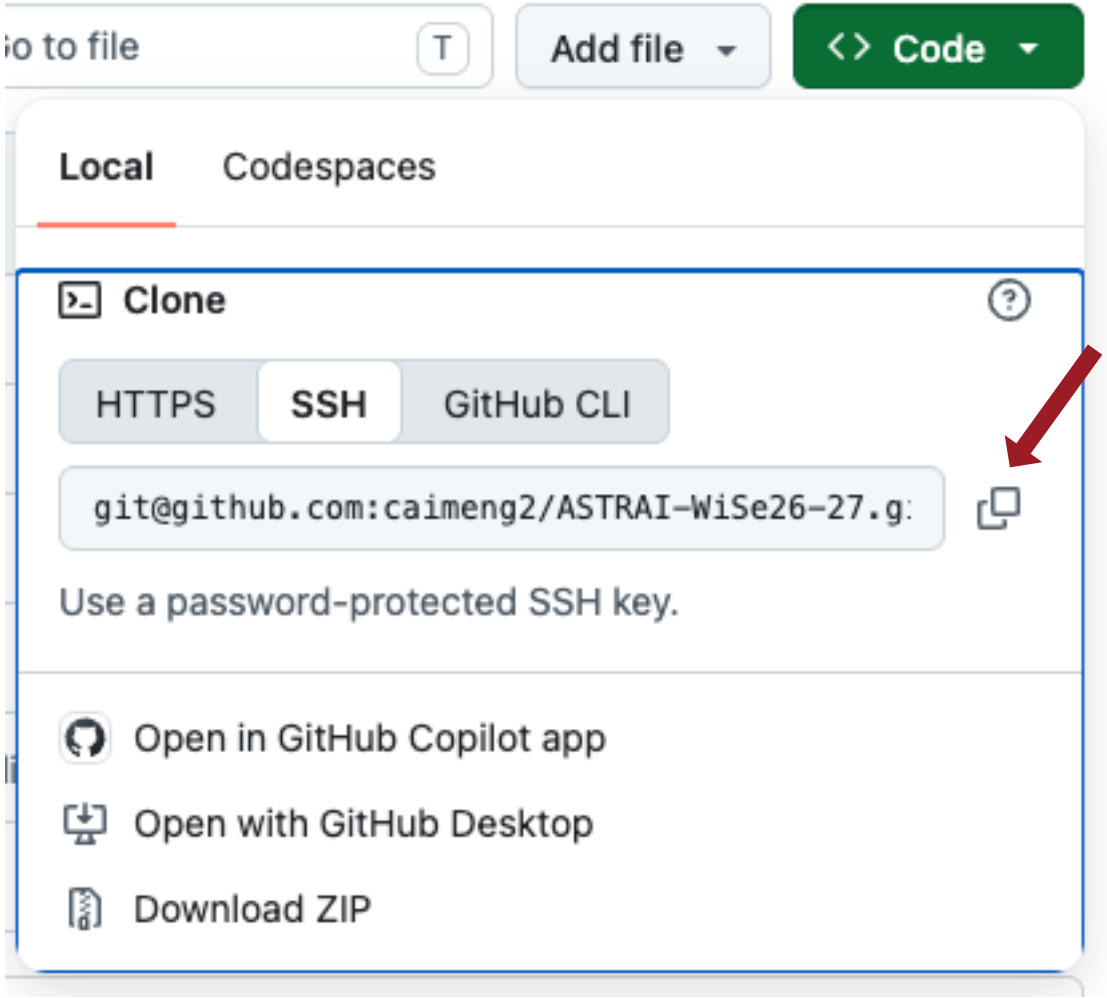
Step 1: Create a folder for this course, for example courses/ASTRAI.

Step 2: Open Terminal or Git Bash in that folder.

- Mac: right-click (two-finger-click) the folder and choose **New Terminal at Folder**. If it isn't there, open Terminal, type cd , drag the folder into the window, and press Enter.
- Windows: right-click inside the folder and choose **Open Git Bash here**. On Windows 11, click Show more options first.

Step 3: On GitHub, click the green <> Code button, then click the copy icon (red arrow).

Step 4: Open Terminal or Git Bash. Type in:
git clone , paste the link, and press Enter



#3.6 Check your cloned ASTRAI repo

You should see a new folder called “ASTRAI-WiSe26-27”

Find the .git and .gitignore. It is hidden by default, so first make hidden files visible:

- Mac: in Finder, press Cmd + Shift + .
- Windows: in File Explorer, go to View → Show → Hidden items

Open .gitignore with a text editor (e.g., TextEdit, Notepad or VS Code) and take a look.

Which files does Git skip here?

.gitignore

```
*.knit.md

# R Environment Variables
.Renviron

# pkgdown site
docs/

# translation temp files
po/*~

# RStudio Connect folder
rsconnect/

# Python
api_key.txt
case_id.txt
.ipynb_checkpoints/

# slides
*.pptx

# word doc
*.docx
*.odt

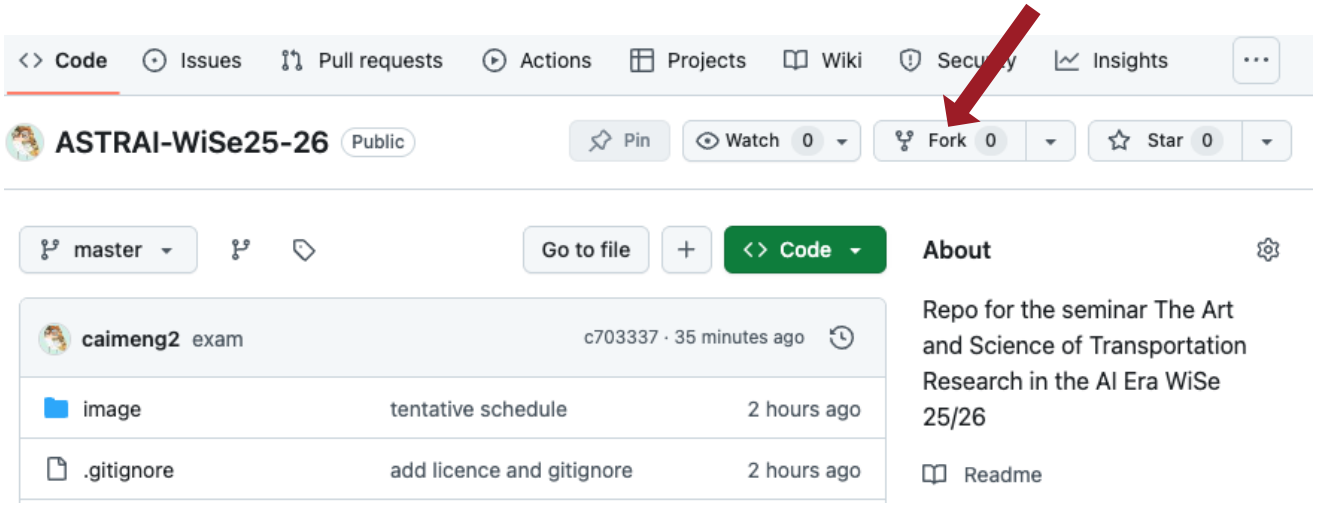
# R
.DS_Store
```

#3.7 Fork ASTRAI repo

Step 1: Sign up for an account at https://github.com

Step 2: Sign in to your account

Step 3: Go to the ASTRAI repo (https://github.com/caimeng2/ASTRAI-WiSe26-27) and click the “Fork” button



#3.8 Pull the changes

I'll now add, commit, and push the slides for this week to the repo.

git pull is the command to use to get updates.

Step 1: Open Terminal or Git Bash inside the ASTRAI-WiSe26-27 folder, the cloned one rather than the folder above it.

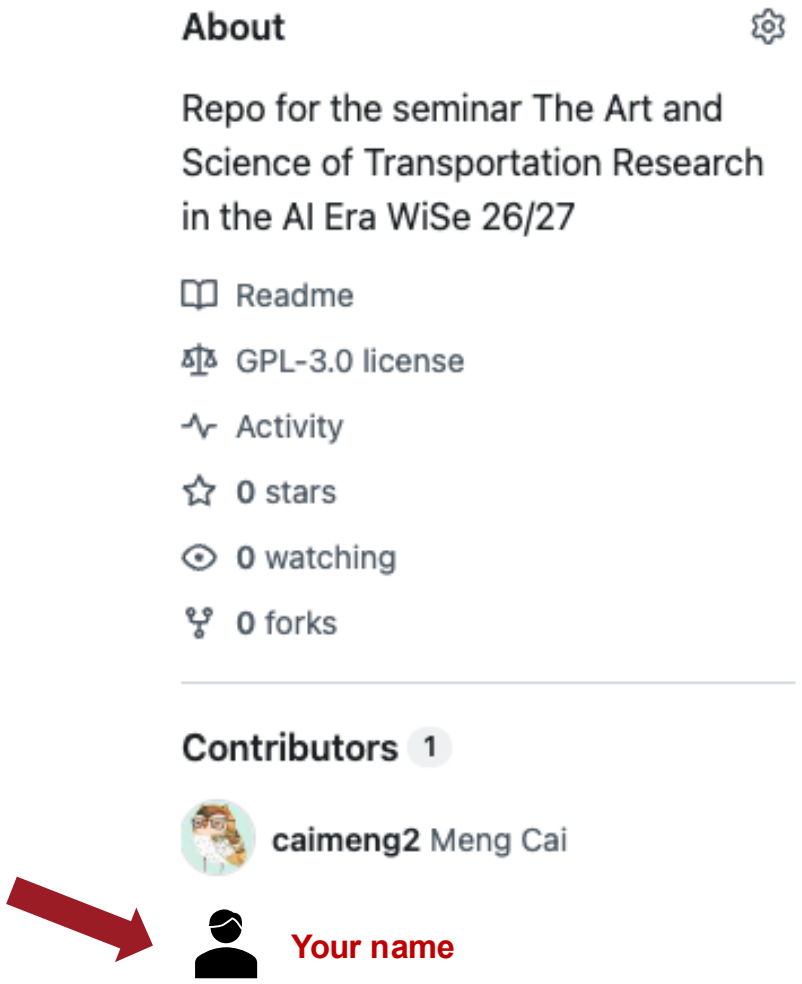
Step 2: Type: git pull

Check your folder. W02_git should appear.

Tip: If you edit the course files directly in the cloned folder, you may run into problems the next time you run git pull. To avoid this, save your edited version **under a new name**, e.g., web_scraping_your_name.ipynb.

#3.9 Again: Why we use Git/GitHub

- Git is an invaluable tool for anyone working with code.
- If you use AI to edit your code, Git shows exactly what changed and lets you undo it.
- Research labs and industry also use Git.
- It keeps course materials open, backed up, and accessible from any device.
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#3.9 Again: git terminology

Remote



The repo in my account

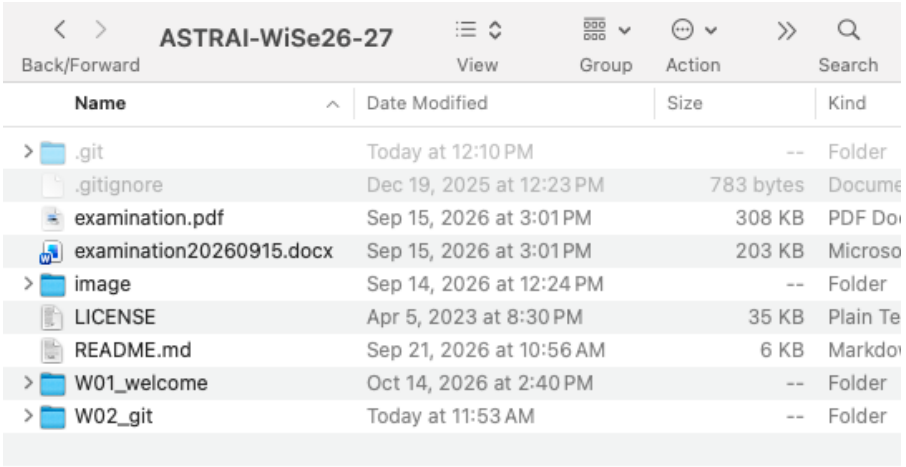




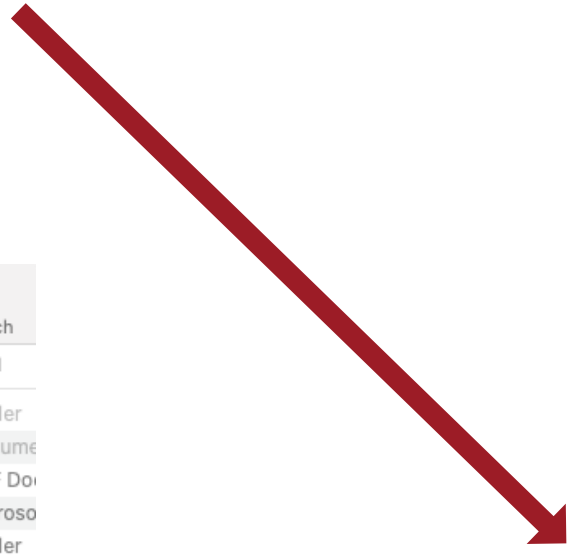
A copy of the repo in your account



Local



The folder on my computer





The folder on your computer

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#4.1 AI reimbursement

About the project: It explores the relationship between university learning and the use of AI.

About the AI reimbursement:

- Claude subscription: I can reimburse your monthly Claude subscription for **up to four months** between now and February 2027.
- Paid APIs: If the method or tool you want to learn and present requires a paid API, I may be able to reimburse that cost as well. Please talk to me first so we can agree on it before you spend anything.
- The total reimbursement is **capped at €100 per person**.

#4.2 Eligibility

Who is eligible:

- You must be enrolled in the course.
- You must take the exam this semester.

#4.3 How it works

1. Sign up today by filling out a survey.

2. Buy Claude yourself and keep all your **invoices, receipts, and proofs of payment**. It must show your name as you registered for this course.

3. During the semester, **learn your chosen method with Claude and share those conversations with me** by pasting them into a Word file. Share only these conversations, not your full chat history. You choose what to include and can remove any passage.

3. Before **25 February 2027**, email me one PDF containing all your invoices, receipts and proofs of payment. In the email, include:
   - Your PayPal or bank details for the transfer
   - Calculate the total amount you are requesting

4. By the end of March 2027, I will send you the money to you by PayPal or bank transfer.

Participation is voluntary and does not affect your grade.

#4.4 Sign up now

https://umfragen.tu-darmstadt.de/325399?lang=en

Please make sure you go to the Google Form at the end:
https://docs.google.com/spreadsheets/d/1dFHLfGnLaVqQtEsdVfxi7d6TQ0-CqqjjuGXuPtnmarI/edit?usp=sharing

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